Animal systems: the evidence challenge

Teacher guide and worked answers

Use evidence about control, circulation and gas exchange to judge claims in an unfamiliar model.

40 minutes. Use Lesson.html for interactive teaching or Lesson.pptx for editable slides. Slides.pdf is the static alternative. Select one pupil response route and print the shared evidence it requires. Detailed answers below are for staff.

Scheme of work

_passsl/inputs/LAUNCH KS4 - 2026-27.xlsx | LAUNCH Weekly - Summer | C34

Apply Topics 7 to 8 to unfamiliar contexts.

Summer C34 application of Topics 7-8. Original integrated practice linked to Pearson 7.13, 8.3-8.5B and 8.12. Fick-style comparisons are Biology-specific, not automatically Higher-only. No diagnosis or complete-topic coverage is claimed.

Prior learning

Insulin helps regulate raised blood glucose.

Cardiac output equals heart rate multiplied by stroke volume.

Area, gradient and distance influence diffusion.

Success evidence

Calculate and compare pump output with units.

Use a controlled comparison to explain a diffusion prediction.

Separate a supported mechanism from a claim the data cannot establish.

Equipment

Shared evidence sheet, one selected response route, pen and ruler.

Projector or offline HTML/PowerPoint; calculator where useful.

Preparation

Print the shared page, one selected response route and the exit page. Routes are alternatives, not extra tasks.

Read the worked answers and note any constructed data before teaching.

Practical care

Paper and screen lesson; no human sampling, exercise test, breath holding, medicine handling or personal health disclosure.

Ready to teach alternative

All evidence is supplied. Use paper diagrams and cards, pointing, annotation or dictation if a device is unavailable.

Teaching sequence

Slide	Stage	Minutes	Focus
1	Opening	0-2	Can the claim survive the evidence?
2	Retrieval	2-5	Recover three mechanisms
3	I do	5-9	Think across the oxygen pathway
4	I do	9-12	Calculate before judging
5	I do	12-15	Keep signal and material separate
6	We do	15-18	Choose a fair comparison
7	We do	18-20	Audit the headline
8	We do	20-22	Test a tempting shortcut
9	Hinge	22-25	Which conclusion is fully supported?
10	You do	25-26	Write a defensible evidence note
11	You do	26-36	Use the data; qualify the claim
12	Review	36-38	Audience: ask for the missing evidence
13	Review	38-39	Make one claim stronger
14	Exit	39-40	Three quick evidence checks
15	Exit	Within stage	Choose the next practice

Times are a suggested 40-minute route. Slides marked Within stage share the preceding stage allocation. The optional HTML timer starts only when selected.

1 Opening Can the claim survive the evidence?

Make clear this is a model audit, not a health test. A pupil can challenge a claim without proposing any treatment.

2 Retrieval Recover three mechanisms

R1 heart rate × stroke volume. R2 down a concentration gradient. R3 a hormone signal; glycogen is a storage carbohydrate.

3 I do Think across the oxygen pathway

Think aloud: “I can use pump output to compare volume per minute. I cannot turn it into oxygen delivery without oxygen content and distribution. I also need exchange and demand to judge adequacy.” Explain aerobic respiration in cells, not in the lungs.

4 I do Calculate before judging

Show beats cancelling in the units. B is 90 × 60 = 5,400 mL/min = 5.4 L/min. A has the larger stroke volume, but B has the greater output in these records.

5 I do Keep signal and material separate

Trace purple signal arrow and blue material arrow. After raised glucose, increased insulin can promote uptake and storage, helping glucose return towards its usual range. Do not claim every cell requires insulin for glucose entry.

6 We do Choose a fair comparison

X/Z changes area alone: predicted index doubles. Y/Z changes thickness alone: Z has half the thickness and twice the index. X/Y changes two factors and cannot isolate either effect.

7 We do Audit the headline

Supported: B pumps a greater volume per minute in these records. Not established: adequacy for every tissue; oxygen content, distribution, exchange and demand were not measured. Ask pupils to name one needed piece of evidence.

8 We do Test a tempting shortcut

X = 3 × 4 / 1 = 12. Y = 6 × 4 / 2 = 12. Both area and thickness double, so effects cancel in this idealised rule. The result is a model prediction, not a clinical observation.

Reveal: X and Y both have index 12. Two factors changed.

9 Hinge Which conclusion is fully supported?

Correct option names the same relative model and unchanged factors. Reteach a comparison by circling the two identical columns.

A. Z has twice X’s relative index because area doubles while concentration difference and thickness stay the same. - The controlled comparison supports that prediction under the supplied rule.

B. B proves that a pupil is fitter than another pupil. - The records are constructed and do not measure pupils, fitness or health.

C. Insulin is broken down into glycogen to provide a store. - Glycogen is made from glucose; insulin is a hormone signal.

10 You do Write a defensible evidence note

The same objective applies across routes. Supply read-aloud or calculator access as needed and record support.

11 You do Use the data; qualify the claim

Supported S1-S4, standard M1-M4 or stretch T1-T4. A correct number without units needs repair; a causal claim needs a comparison or mechanism.

12 Review Audience: ask for the missing evidence

A useful next measure could be stroke volume under a defined condition, oxygen content, tissue blood distribution or a controlled exchange variable. Accept a focused question, not a vague “more data”.

13 Review Make one claim stronger

Pupil records one revision. Ask whether the new evidence could distinguish two explanations.

14 Exit Three quick evidence checks

E1 4 L/min. E2 area and thickness both change. E3 demand, oxygen content, distribution or actual exchange. No medical interpretation is needed.

15 Exit Choose the next practice

Shared exit minute; let pupils point to a next step and explain briefly.

Answers and assessment

Retrieval

R1 cardiac output = heart rate × stroke volume. R2 net diffusion down a concentration gradient. R3 insulin is a hormone signal; glycogen is a store made from glucose.

We do and hinge

W1 X/Z varies area only; Z has twice X's index. W2 Y/Z varies thickness only; Z has half Y's thickness and twice its index. W3 B has greater volume output, but adequacy is unproven. X and Y both give index 12 because area and thickness both double. First hinge option is correct; the other options falsely infer pupil fitness or turn insulin into glycogen.

S1 / M1

A = 60 × 70 = 4,200 mL/min = 4.2 L/min. B = 90 × 60 = 5,400 mL/min = 5.4 L/min. Difference = 1.2 L/min. B is greater. Heart rate alone ignores stroke volume.

S2 / M2

Area-only pair X/Z: concentration difference 4 and thickness 1 are unchanged. X = 12, Y = 12, Z = 24. X/Y changes area and thickness together, so it cannot isolate one factor. Index units are relative, not calibrated physiological units.

S3 / M3

Raised glucose stimulates pancreatic beta cells to release insulin. Insulin signals responsive tissues and promotes uptake and glycogen storage in liver/muscle; blood glucose moves towards its usual range. Glucose is a transported material; glycogen is its storage carbohydrate. Some uptake is insulin-independent.

S4 / M4

Ventilation replenishes alveolar air; oxygen diffuses into pulmonary blood, is carried by circulation to tissue capillaries and diffuses towards cells, where it supports aerobic respiration. Pump output alone omits oxygen content, distribution, exchange performance and tissue demand. Any two explained limitations are valid.

T1

C = 120 × 35 = 4,200 mL/min = 4.2 L/min, equal to A and below B. C has the highest heart rate but not the highest output. Both factors matter.

T2

Keep concentration difference 4 and thickness 1; use area 9. Predicted index = 9 × 4 / 1 = 36, three times X. Other assumptions and the proportionality constant must also match.

T3

At the same output, less oxygen per unit blood could reduce oxygen delivery. Greater tissue oxygen demand could make the same delivery insufficient. These are conditional mechanisms; the supplied data do not measure either quantity or identify a person's condition.

T4

Example: B has greater volume output than A, 5.4 versus 4.2 L/min. This does not establish adequate oxygen supply to every tissue. Measure oxygen content of the supplied blood and specify tissue demand to test that stronger claim. Accept another coherent conclusion/limit/measurement trio.

Exit and success

E1 80 × 50 = 4,000 mL/min = 4 L/min. E2 both area and thickness change. E3 oxygen content, distribution, exchange or tissue demand. Secure work combines an accurate calculation, a mechanism and a specific limit; reteach whichever link is missing.

L1

Accept a recorded evidence question and a meaningful revision, such as adding “under this model” or replacing an unsupported health conclusion with a specific output comparison.

Teaching and assessment notes

40-minute lesson. Zero-minute slides share the preceding stage allocation.

Choose one response route. Accept accurate speech, pointing with explanation, annotations or writing; record the support used.

A teacher or TA can be a quiet audience: ask one evidence question, then let the pupil revise or justify a decision.

Interactive prediction-test-explain uses the same pump records and X/Y/Z cases as the printable sheet. Pupils can point to columns and calculate on paper without losing the scientific decision.

Do not invite diagnoses, personal fitness comparisons or medication advice. These are constructed teaching records with explicitly limited claims.

Optional stretch is about evidence and model design. Glucagon and other Higher-only endocrine mechanisms are not needed.

Access and responsive teaching

Supported

Same core concepts with chunked prompts and read-aloud access. Record support used; a pathway is not a diagnosis or fixed attainment label.

Standard

Explain the supplied evidence independently using accurate scientific terms and units.

Stretch

Optional deeper transfer or evaluation. Higher-tier-only material is labelled explicitly and is not required for the core outcome.

Regulation

Offer solo or partner work, a quiet response, a pass and brief reset with clear re-entry. No forced disclosure or public performance.

Misconceptions to check

A faster heart rate always means greater cardiac output.

Stroke volume also matters; multiply both quantities.

Check: Compare both columns before calculating.

Twice the area always doubles diffusion rate.

That prediction requires the other relevant factors to be held constant.

Check: Compare X with Y, where thickness also changes.

A pump-output number proves that a tissue receives enough oxygen.

Adequacy also depends on oxygen content, distribution, exchange and tissue demand.

Check: Ask which of those quantities was measured.

Insulin is the fuel carried to cells.

Glucose is a fuel; insulin is a signal that affects target-tissue responses.

Check: Label one signal arrow and one material arrow.

Word	Meaning
cardiac output	The volume of blood pumped by one ventricle per unit time.
stroke volume	The volume of blood pumped by one ventricle per beat.
controlled comparison	A comparison in which the factor of interest changes while relevant other factors are kept the same.
model limitation	Something a simplified representation does not include or cannot establish.
insulin	A pancreatic hormone involved in reducing raised blood glucose.
glycogen	A storage carbohydrate made from glucose.

Scientific references

Pearson Edexcel GCSE Biology 1BI0 specification

Used to check relevant Foundation curriculum content; this is teacher-authored practice, not an official assessment or complete unit. Checked 9 September 2026.

NHLBI: How blood flows through the heart

Used for the circulation and oxygen-transport pathway. Checked 9 September 2026.

NHLBI: What breathing does for the body

Used for alveolar exchange and oxygen transport. Checked 9 September 2026.

NIDDK: What is diabetes?

Used for insulin, glucose and energy context. All numerical records are constructed classroom data, not patient measurements. Checked 9 September 2026.